mdcD (C5APL4) — Hypothesis Review: "acetyl-CoA carboxylase activity (GO:0003989)"

Gene: mdcD · **Organism:** *Methylorubrum extorquens* AM1 (NCBITaxon:272630) · **UniProt:** C5APL4 (310 aa) **Focus:** function_assignment · **Seed hypothesis:** *mdcD has acetyl-CoA carboxylase activity (GO:0003989)* · **Evidence type in review:** IEA (GO_REF:0000118)

Executive Judgment

Verdict: REFUTED / over-annotated.

`mdcD` encodes the **β-subunit of a biotin-independent malonate decarboxylase (MDC)**, not an acetyl-CoA carboxylase. The GO:0003989 annotation is a phylogenetic over-annotation produced by TreeGrafter (GO_REF:0000118) grafting the protein to a PANTHER *family* node (PTHR42995, "acetyl-CoA carboxylase carboxyl transferase subunit β"), while the matching PANTHER *subfamily* (PTHR42995:**SF1** = **"malonate decarboxylase beta subunit"**) and the malonate-specific HMM signatures (NCBIfam **TIGR03133 "malonate_beta"**, InterPro **IPR017556 "Malonate_beta"**) correctly identify the protein.

The acetyl-CoA carboxylase reaction (GO:0003989 definition: `ATP + acetyl-CoA + HCO₃⁻ → ADP + Pi + malonyl-CoA`) is a **biotin-dependent, multi-step carboxylation** requiring biotin-carboxylase (ATP-grasp) and biotin-carboxyl-carrier (BCCP) functions. MdcD contains **only** a single carboxyltransferase domain (Pfam **PF01039 Carboxyl_trans**, InterPro IPR011762 COA_CT_N; crotonase/ClpP-like fold) and has **no ATP-grasp and no BCCP domain**, so it is mechanistically incapable of the GO:0003989 reaction. The referenced crystal-structure paper

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shows directly that MdcD–MdcE form the **decarboxylase** catalytic component of the MDC holoenzyme and that a CoA thioester binds the MdcD–MdcE active site — a decarboxylation, not an acetyl-CoA carboxylation, chemistry.

Organism-specific corroboration (Iteration 2): KEGG annotates the exact target gene **mea:Mex_1p0025 (= C5APL4)** as "malonate decarboxylase, beta-subunit", KEGG Orthology **K13932, EC 4.1.1.87** (*malonyl-[malonate decarboxylase] decarboxylase*, a class-4 carboxy-lyase) — a different enzyme class from acetyl-CoA carboxylase (EC 6.4.1.2, class 6). A UniProt proteome search of *M. extorquens* AM1 recovers the **intact, syntenic mdc operon** (mdcA C5APL2, mdcB C5APL7, mdcC C5APL3 [ACP], mdcD C5APL4, mdcE C5APL5), consistent with a complete, functional biotin-independent malonate decarboxylase in this organism rather than a fatty-acid-biosynthetic ACC.

Caveats: (i) all target annotations are IEA/electronic — there is no experimental characterization of the *M. extorquens* AM1 protein itself; (ii) the strongest experimental evidence

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is from a *Pseudomonas* ortholog; (iii) GO lacks a dedicated "malonate decarboxylase activity" MF term, so the best supported replacements are general (carboxy-lyase / carboxyltransferase).

Evidence Matrix

Citation	Evidence type	Direction	Claim tested	Key finding	Context
P 28757619	Structural (X-ray) + mutagenesis	**Refutes**	Does MdcD carboxylate acetyl-CoA?	MdcD–MdcE are the **decarboxylase** catalytic subunits of biotin-independent MDC; CoA bound in the MdcD–MdcE active site	*Pseudon* MDC holoenz (ortholo
UniProt C5APL4 (TIGR03133 / IPR017556 / PANTHER SF1)	Computational / domain	**Refutes → reassigns**	Protein identity & function	Classified as **malonate decarboxylase β subunit**; single Carboxyl_trans (PF01039) domain; **no biotin-carboxylase / BCCP**	*M. extorque* AM1 (ta
GO:0003989 definition (QuickGO)	Ontology / database	**Refutes**	Can a lone carboxyltransferase run the ACC reaction?	Term = `ATP + acetyl-CoA + HCO₃⁻ → ADP + Pi + malonyl-CoA`; biotin-dependent, multi-domain — impossible for MdcD alone	N/A
UniProt GO set (C5APL4)	Database (IEA)	**Qualifies**	Reliability of electronic annotations	Protein carries several **conflicting** IEA MF terms (GO:0003989 ACC; GO:0004658 propionyl-CoA carboxylase; GO:0016831 carboxy-lyase) — hallmark	Target

Citation	Evidence type	Direction	Claim tested	Key finding	Context
				carboxyltransferase carry-over	
TreeGrafter / GO_REF:0000118	Computational (phylogenetic)	**Competing**	Provenance of GO:0003989	Grafted at **family** node PTHR42995 (ACC CT-β); **subfamily SF1** = **malonate decarboxylase β** is more specific and correct	N/A
KEGG mea:Mex_1p0025 (KO K13932)	Database	**Refutes → reassigns**	Organism-specific function of mdcD	Named "malonate decarboxylase, beta-subunit"; **K13932, EC 4.1.1.87** (carboxy-lyase, class 4) — *not* EC 6.4.1.2; only Carboxyl_trans motif	*M. extorque* AM1 (ta
UniProt proteome search (taxon 272630)	Database / evolutionary	**Supports reassignment**	Is the full MDC operon present in METEA?	Intact, syntenic **mdc cluster**: mdcA (C5APL2), mdcB (C5APL7), mdcC (C5APL3, ACP), mdcD (C5APL4), mdcE (C5APL5)	*M. extorque* AM1 (ta

Provenance artifacts: `provenance/mdcD_evidence_matrix.csv`, `provenance/mdcD_GO_decision_table.csv` (UniProt/QuickGO queries executed live via the REST APIs).

GO Curation Implications (leads — require curator verification)

GO ID	Label	Aspect	Action	Rationale
GO:0003989	acetyl-CoA carboxylase activity	MF	**REMOVE**	Over-annotation; MdcD cannot catalyze the biotin-dependent ACC reaction
GO:0004658	propionyl-CoA carboxylase activity	MF	REMOVE	Same carboxyltransferase-domain carry-over (note: also present on C5APL4)
GO:0006633	fatty acid biosynthetic process	BP	REMOVE/ REVISE	Downstream of the ACC mis-assignment; MdcD is catabolic
GO:2001295	malonyl-CoA biosynthetic process	BP	REMOVE/ REVISE	Downstream of the ACC mis-assignment
GO:0016831	carboxy-lyase activity	MF	**RETAIN**	Correct, mechanism-consistent general class (already IEA/InterPro)
GO:0016743	carboxyl- or carbamoyltransferase activity	MF	ADD (lead)	Reflects the MdcD–MdcE carboxyltransferase step in the holoenzyme
GO:0090410	malonate catabolic process	BP	ADD (lead)	Correct biological process for the mdc operon

No dedicated "malonate decarboxylase activity" MF term exists in GO; `carboxy-lyase activity` (GO:0016831) is the most informative supported MF. "Protein binding" is **not** recommended.

Sequence Discriminator (Iteration 3, computed)

A Needleman–Wunsch global alignment (own code; provenance `provenance/mdcD_sequence_identity_comparison.csv`) places mdcD (C5APL4) with malonate decarboxylase, not with acetyl-CoA carboxylase:

C5APL4 vs	Role of reference	Global % identity
Q9I6S7 (*P. aeruginosa* MdcD)	malonate decarboxylase β-subunit (structure organism)	**42.5 %**
P0A9Q5 (*E. coli* AccD)	**bona fide** ACC carboxyltransferase β	18.3 %
P0ABD5 (*E. coli* AccA)	**bona fide** ACC carboxyltransferase α	22.1 %

mdcD is ~2× more similar to a malonate decarboxylase β-subunit than to any genuine acetyl-CoA carboxylase subunit; the residual 18–22 % reflects only the shared carboxyltransferase fold — exactly the signal that misleads automated (family-level) annotation.

Mechanistic Scope

- **Direct molecular function tested:** whether MdcD carboxylates acetyl-CoA (ATP-/biotin-dependent). **Not supported.**
- **Actual direct function:** MdcD is the β carboxyltransferase subunit that, together with MdcE (γ), forms the **decarboxylase catalytic module** of the malonate decarboxylase holoenzyme (also containing MdcC acyl-carrier protein and MdcA ACP-transferase). The module decarboxylates a malonyl-thioester (malonyl-S-ACP) to acetyl-S-ACP + CO_2.
- **Pathway consequence (not direct activity):** malonate catabolism, allowing malonate use as a carbon/energy source — a *downstream* process, not the per-subunit molecular activity.
- The seed term conflates a shared **structural fold** (carboxyltransferase, common to ACC-β, PCC, methylmalonyl-CoA decarboxylase, and MDC-β) with a specific **reaction**.

Conflicts and Alternatives

- **Paralog/family confusion (primary driver):** The PF01039 carboxyltransferase fold is shared across biotin-dependent carboxylases (ACC, propionyl-CoA carboxylase) and biotin-independent decarboxylases (MDC). Automated pipelines inherit the family label (acetyl-CoA carboxylase) instead of the subfamily (malonate decarboxylase β). The same protein simultaneously carries GO:0004658 (propionyl-CoA carboxylase) — an internal contradiction confirming annotation noise.
- **Organism specificity:** Experimental proof is from *Pseudomonas* (P 28757619) and historically *Klebsiella/Acinetobacter/Malonomonas*; the *M. extorquens* AM1 protein is inferred by orthology/synteny (mdc operon) and is not individually assayed.
- **No competing experimental claim** supports an ACC role for any MdcD ortholog.

Knowledge Gaps

1. **No direct biochemistry on C5APL4 itself.** Checked: UniProt shows only IEA. Matters because curation ideally wants organism-specific support. Resolved by: in-vitro reconstitution/decarboxylation assay of the *M. extorquens* MDC, or demonstrated malonate-dependent growth phenotype on knockout.
2. **mdc operon completeness / synteny in METEA.** Checked: single-protein UniProt record only; did not enumerate neighboring mdcA/C/E/G/H. Matters because the holoenzyme context strengthens the decarboxylase call. Resolved by: genome-neighborhood analysis (e.g., NCBI/KEGG operon view).
3. **Exact GO MF granularity.** Checked QuickGO: no "malonate decarboxylase activity" term exists. Matters for choosing the replacement. Resolved by: proposing a new GO term or using GO:0016743/GO:0016831 with the holoenzyme noted in annotation extensions.

Discriminating Tests

- **Domain/sequence:** confirm absence of ATP-grasp (PF00289 Biotin_carb) and BCCP (PF00364) in C5APL4 — already indicated by UniProt; a dedicated HMMER scan would formalize it.
- **Genomic context:** verify the mdcABCDE(GH) operon neighbors in *M. extorquens* AM1.
- **Biochemical:** malonate decarboxylase assay (CO_2/acetate release from malonyl-thioester) on reconstituted METEA subunits; ACC assay (malonyl-CoA formation from acetyl-CoA + bicarbonate + ATP) as the **negative** discriminator — expected inactive.
- **Genetics:** mdcD deletion → loss of growth on malonate as sole carbon source (parallels the *P. aeruginosa* result in

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Curation Leads (require curator verification)

- **Action:** Remove GO:0003989 (and flag GO:0004658, GO:0006633, GO:2001295 as derived over-annotations). Retain GO:0016831. Consider adding GO:0016743 (MF) and GO:0090410 (BP), annotating MdcD as the β carboxyltransferase subunit of the malonate decarboxylase holoenzyme.
- **Candidate reference to cite:**

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  — verify snippet: *"...decarboxylase (MdcD-MdcE) and acyl-carrier protein transferase (MdcA) catalytic activities."* and *"...the structure of malonate decarboxylase with CoA in the active site of MdcD-MdcE."*
- **Supporting signatures to cite:** UniProt C5APL4 cross-refs TIGR03133 (malonate_beta), InterPro IPR017556 (Malonate_beta), PANTHER PTHR42995:SF1 (malonate decarboxylase beta subunit).
- **Suggested questions for curator:** Is there a METEA-specific malonate-utilization phenotype or operon annotation? Should the review also down-rank the co-occurring propionyl-CoA carboxylase IEA term?
- **Suggested experiment:** reconstituted MDC decarboxylase assay + mdcD knockout growth test on malonate.

Limitations

Evidence is orthology- and domain-based for the target protein; the decisive functional data are from a *Pseudomonas* ortholog. No local bioinformatics files were available in this run; all computational checks used live UniProt and QuickGO REST queries.